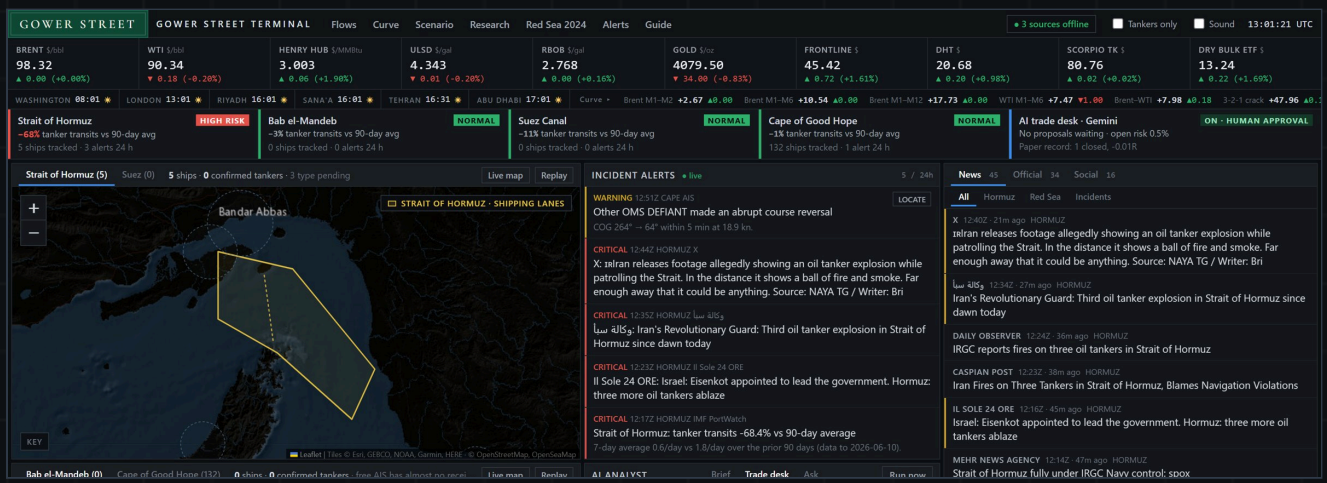
GOWER STREET

# Getting started on Gower Street Terminal.

A real-time maritime-risk terminal for the world's oil chokepoints: the Strait of Hormuz, Bab el-Mandeb, the Suez Canal and the Cape of Good Hope.






Mohammadamin Zare · University College London

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## How to use this guide

Read sections 1 and 2 for a five-minute orientation. Each later section covers one part of the screen: what it shows, where the data comes from and what you can do with it. Section 12 has a one-page quick reference.

> **Open the terminal:** gower-street-terminal-sg.onrender.com. It runs in any modern browser on a computer, tablet or phone; no login or installation is needed. On free hosting, the first visit after a quiet spell can take up to a minute to wake the server.

01 · ABOUT THE TERMINAL

# What Gower Street Terminal does

About a fifth of the world's oil passes through a handful of narrow sea lanes. When a tanker is attacked, seized or goes dark in one of them, prices move within minutes. Gower Street Terminal puts everything a trader needs to judge that moment on one screen, and helps decide what, if anything, to do about it.

## Four chokepoints, one screen

| Chokepoint | Why it matters |
|---|---|
| **Strait of Hormuz** | The exit from the Persian Gulf; the single most important oil route in the world. |
| **Bab el-Mandeb** | The southern gate of the Red Sea; the target of the 2023–24 attacks on shipping. |
| **Suez Canal** | Links the Red Sea and the Mediterranean; traffic falls when Bab el-Mandeb is unsafe. |
| **Cape of Good Hope** | The long way round Africa; traffic rises when ships avoid the Red Sea. |

## What it brings together

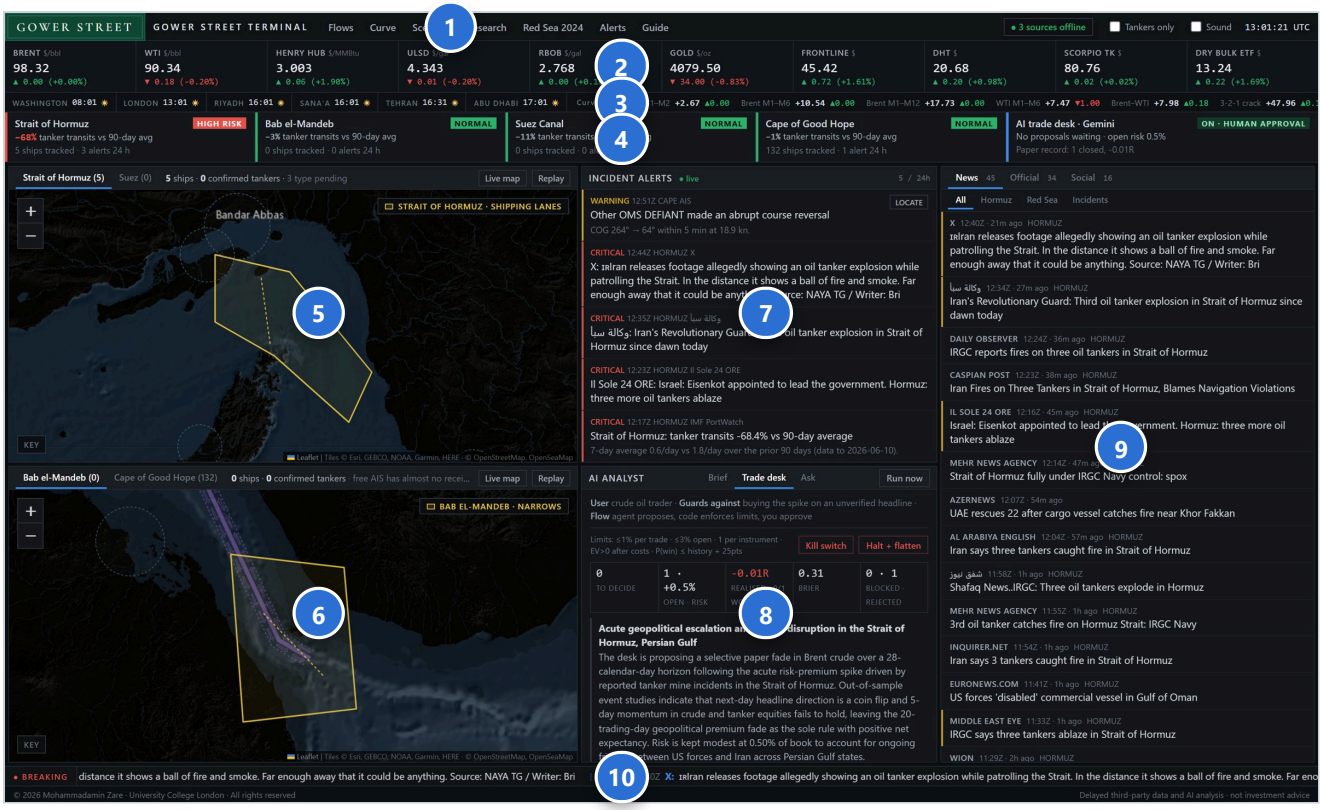
- **Live ships** from AIS transponder data, with automatic alerts for ships going dark, GPS jamming, sanctioned tankers and ship-to-ship transfers.
- **Energy prices** (Brent, WTI, gas, products, gold and tanker shares) and futures spreads.
- **Credited news and official statements** from every side: Iran, Israel, the Houthis, the Persian Gulf states, the UK, the UN and the US.
- **IMF PortWatch transit data** showing how many ships really pass each chokepoint per day.
- **An AI analyst and trade desk** powered by Google Gemini, which proposes paper trades that code prices, checks and logs, and that a human must approve.
- **Research** testing the ideas on 20 past incidents since 2019, out of sample and after costs.

> **Paper trading only.** No part of the terminal can place a real order. Prices are delayed, AIS coverage has gaps, and nothing here is investment advice.

02 · THE SCREEN AT A GLANCE

# The screen at a glance

Everything updates live; you never need to refresh. On a computer the whole terminal fits one screen. On a phone the panels stack and you scroll.




1. **Top bar:** logo, tools menu, data status, Tankers only, Sound, UTC clock.
2. **Prices:** live (delayed) energy, gold and tanker-share prices. Click one for its chart.
3. **Clocks and spreads:** local times in key capitals; futures spreads that signal supply stress.
4. **Risk banner:** a risk level for each chokepoint, and the trade desk's status.
5. **Map: Hormuz / Suez.** Live ships and watched areas.
6. **Map: Bab el-Mandeb / Cape.** The second waterway pair.
7. **Incident alerts:** everything the terminal detected, newest first.
8. **AI analyst:** the trade desk, the written brief and the Ask chat.
9. **Feeds:** credited news, official statements and social posts.
10. **Breaking strip:** the latest critical items scrolling across the bottom.

03 · TOP BAR, PRICES AND CLOCKS

# Top bar, prices and clocks



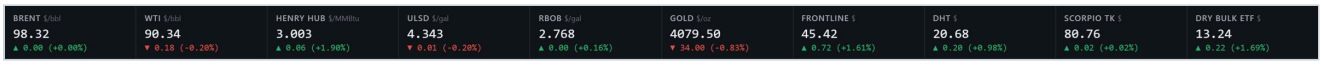

The top bar: logo, tools menu, data status and display options.

## Tools menu

| Button | Opens |
|---|---|
| **Flows** | Daily ships through each chokepoint (IMF PortWatch) against the terminal's own counts. |
| **Curve** | Brent and WTI futures curves and spreads, with what each one tells you. |
| **Scenario** | A what-if calculator: how much oil a disruption removes and what it could do to prices. |
| **Research** | The event study and out-of-sample tests on 20 past incidents. |
| **Red Sea 2024** | A day-by-day replay of the 2023–24 Red Sea crisis. |
| **Alerts** | Phone alerts (ntfy or Telegram) and the status of every data source. |
| **Guide** | A 30-second tour of the screen. It also runs once on your first visit. |

On the right: ● **Data** shows which sources are live; **Tankers only** hides every other ship on the maps; **Sound** beeps on new alerts.

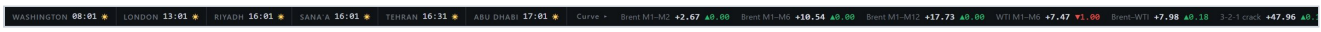
## Prices




Each tile shows the last price, the change and the percentage change. Prices are delayed third-party quotes.

Click any price for a full chart with candles or a line, several ranges and markers for past incidents.

## Clocks and spreads



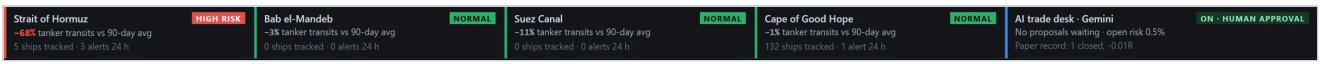

Local time in Abu Dhabi, London, Riyadh, Sana'a, Tehran and Washington; then futures spreads in $/bbl.

A **positive time spread** (backwardation) means oil for delivery now costs more than oil later: a sign the market fears a shortage. Click the spreads for the full curve.

04 · CHOKEPOINT RISK BANNER

# Chokepoint risk banner

The banner answers the first question anyone asks: is anything wrong right now, and where?




One tile per chokepoint, then the AI trade desk tile on the right.

## Reading a tile

- **Risk level** in the corner: NORMAL ELEVATED HIGH RISK.
- **Tanker transits** over the last 7 days against the 90-day average, from IMF PortWatch.
- **Ships tracked** right now and the number of **alerts** in the last 24 hours.

Hover over a tile to see exactly why it has its level. Click it to jump the map to that waterway.

## How the level is set

The level comes from a simple, published rule, not from the AI, so anyone can check it:

| Level | When |
|---|---|
| HIGH RISK | Tanker transits at least 40% below normal, or a critical alert in the last 24 hours. |
| ELEVATED | Transits 15–40% below normal, three or more warnings in 24 hours, or a sanctioned ship in the area. |
| NORMAL | None of the above. |

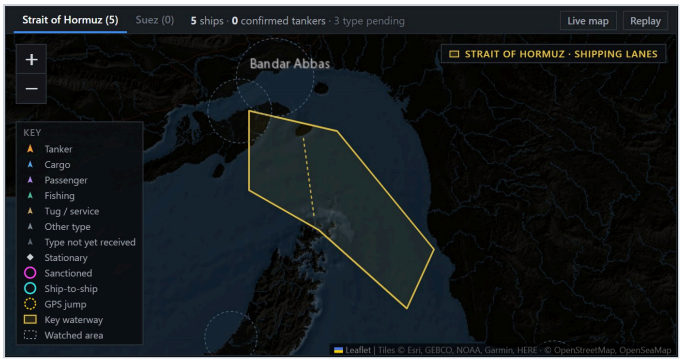
## The trade desk tile

Shows whether the desk is on and under human approval (or halted by the kill switch), how many proposals are **awaiting your approval**, the open risk and the paper record. Click it to open the trade desk.

> **Note:** IMF PortWatch publishes transits with a delay. The hover text always gives the date the data runs to.

05 · SHIP MAPS

# Ship maps




The Strait of Hormuz on the terminal's own map, with the key open.

### Two panels, four waterways

The top panel shows the Strait of Hormuz or Suez; the bottom one Bab el-Mandeb or the Cape. Switch with the tabs in each panel's header, which also show how many ships are tracked.

### Own map and Live map

**Own map** draws the terminal's own AIS data, so every ship on it feeds the alerts. **Live map** switches to VesselFinder's map, which has wider coverage but is their data, not used by the alerts.

### Replay

Plays back the last 24 hours of ship movement with a time slider.

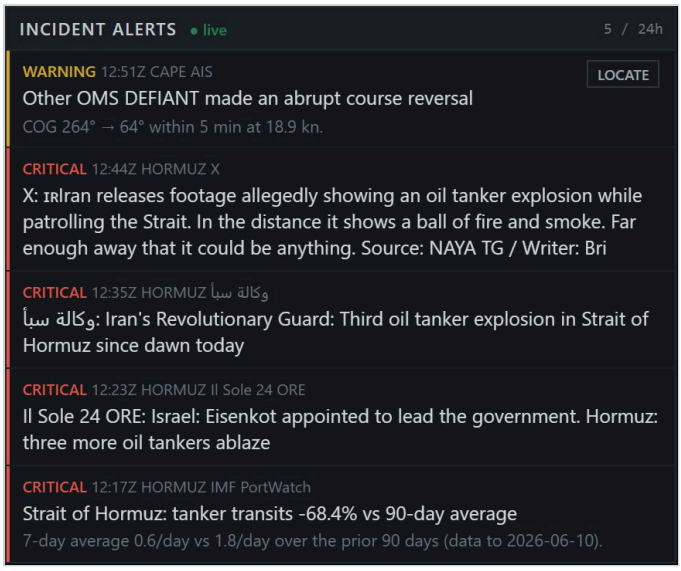
## The key

| Symbol | Meaning |
|---|---|
| **Arrow (larger, orange)** | Tanker, pointing the way it is heading. Names appear when you zoom in. |
| **Arrows by colour** | Cargo, passenger, fishing, tug or service, other, or type not yet received. |
| **Diamond** | Stationary ship. |
| **Magenta ring** | On the US Treasury OFAC sanctions list. |
| **Cyan ring** | Possible ship-to-ship transfer. |
| **Yellow dashed ring** | GPS position jump (possible jamming or spoofing). |
| **Yellow box** | The watched waterway; the dashed line is the gate used to count transits. |

> **Tip:** zoom with the mouse wheel or the + and − buttons, and tick **Tankers only** in the top bar to hide everything else. Free AIS relies on shore receivers, so coverage inside the straits is thin; that gap is itself shown, not hidden.

06 · INCIDENT ALERTS

# Incident alerts




Newest first. Red edge = critical, amber = warning.

The terminal watches every ship, price, spread and feed, and raises an alert when something changes. Each alert shows its time (UTC), waterway, source and what was detected.

Click **LOCATE** to fly the map to the ship. The counter in the header shows how many alerts arrived in the last 24 hours.

Critical alerts also scroll across the breaking strip at the bottom of the screen and can be sent to your phone (section 12).

## What triggers an alert

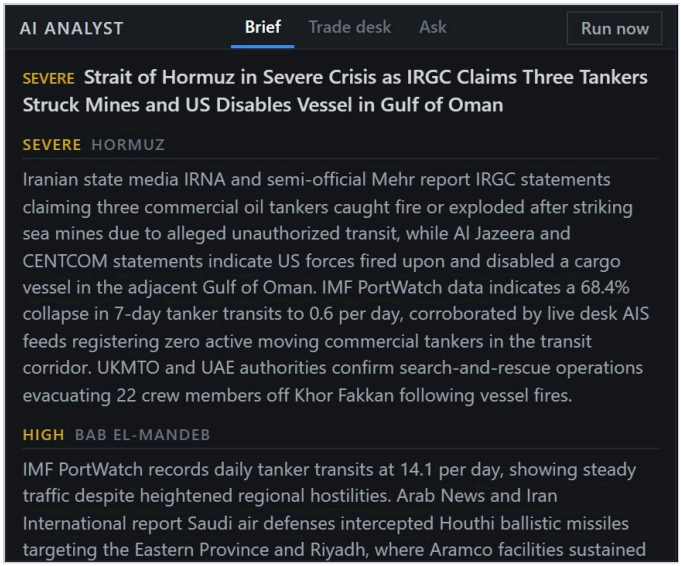
| Alert | Detected when |
|---|---|
| **AIS dark** | A tanker under way stops transmitting inside a watched area. |
| **GPS jump / interference** | A ship's position leaps impossibly far; several at once means likely jamming. |
| **Sanctioned vessel** | A ship matching the OFAC sanctions list appears. |
| **Ship-to-ship transfer** | Two ships sit side by side at sea, away from an anchorage. |
| **Sudden stop / course reversal** | A ship stops abruptly at sea, or turns back sharply at speed. |
| **Navigation emergency** | A ship reports "not under command" or a similar status. |
| **Transit drop** | IMF PortWatch transits fall well below their 90-day average. |
| **Price / spread move** | A price moves 1% or more (critical at 4%), or a futures spread jumps. |
| **News / official incident** | A credited outlet or official source reports an incident in a watched area. |

> **Detections are leads, not facts.** A ship can go dark because of a coverage gap; a GPS jump can be a faulty transponder. Check the details before acting.

07 · AI ANALYST

# AI analyst: Brief and Ask

The AI analyst panel has three tabs: **Trade desk** (section 8), **Brief** and **Ask**. All three run on Google Gemini.




The Brief tab: a written read of the live picture.

### Brief

Gemini reads the live ship statistics, alerts, credited headlines, official statements, prices, the futures curve and IMF transit data, and writes a short assessment for each waterway, with a severity and the evidence behind it.

It refreshes on a schedule and after major alerts. **Run now** asks for a fresh one.

### Ask

Ask anything about the live picture in plain English, for example "*Why did Brent drop at 14:00?*" or "*Which tankers went dark in Hormuz today?*". Answers come only from the terminal's current data.

On the public site each visitor has a small daily number of questions; the counter shows how many are left.

> **Why the AI never sets prices:** the analyst may judge what an event means, but every number on the trade desk (levels, costs, expected value, risk) is computed in code. That way a model error cannot become a wrong price.

08 · THE AI TRADE DESK

# The AI trade desk

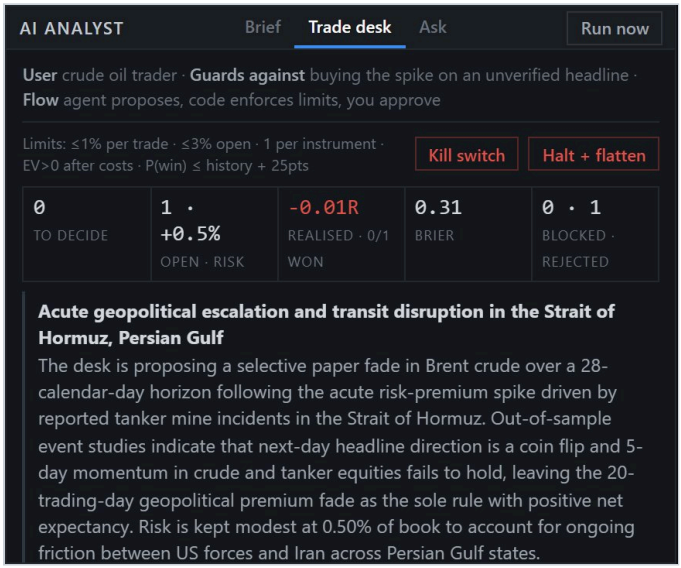
The desk turns events into paper trades a portfolio manager could defend: the AI proposes, code checks every number, and a human decides.

STEP 1
**Gemini proposes**
An instrument, a direction, a horizon, a probability, a historical analogue and its sources.

STEP 2
**Code prices it**
Entry, stop, target, round-trip cost and expected value, with the formula shown.

STEP 3
**Code checks it**
Every hard risk limit below. A trade that fails any one is blocked.

STEP 4
**You decide**
Approve or reject. Nothing opens without a person, and every action is logged.




The trade desk: limits, kill switch, scoreboard and the desk's current view.

### Hard risk limits

| | |
|---|---|
| **Per trade** | at most 1% of book at risk |
| **Open risk** | at most 3% of book in total |
| **Per risk group** | at most 2% in crude, tankers, gas or gold |
| **Per instrument** | one position at a time |
| **Expected value** | positive after costs |
| **Confidence** | at most 25 points above history |
| **Stop distance** | at least ½ a normal move |
| **Evidence** | at least one cited source |
| **Expiry** | unapproved proposals lapse after 60 min |

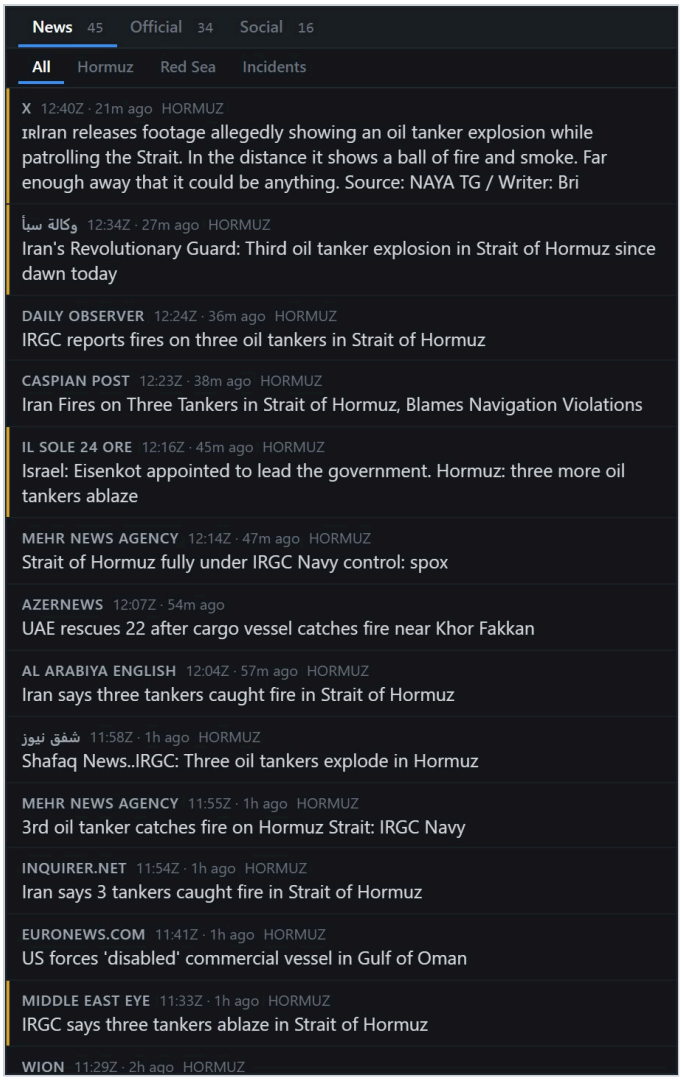
## Controls

- **Approve / Reject** on each proposal; **Close** on an open trade.
- **Kill switch** halts all new trading; **Halt + flatten** also closes everything open.
- **Decision trace** links each trade to the exact data the AI saw; the **audit log** records every action by the AI, the server or a person.

Results are scored in R (multiples of the risk taken) after costs, with a Brier score for how well-calibrated the probabilities were.

09 · NEWS, OFFICIAL STATEMENTS AND SOCIAL

# News, official statements and social




Every item names its source and links to the original.

### Three feeds

**News** collects headlines from named outlets and Google News. **Official** collects what governments and armed groups say themselves. **Social** follows selected public accounts and channels. The number on each tab is how many items it holds.

### Filters

News filters: **All**, **Hormuz**, **Red Sea**, **Incidents**.

Official filters, one per side: **All**, **Houthi**, **Iran**, **Israel**, **Persian Gulf** states, **UK**, **UN** and **US**.

### Reading an item

The source, time (UTC), age and waterway sit above each headline. An amber edge marks an incident. Click a headline to open the original article.

> **Why every side?** In a crisis each government tells its own story. Seeing Iran, Israel, the Houthis, the Persian Gulf states, the UK, the UN and the US side by side shows where they agree and where they don't.

## Breaking strip




Critical alerts, official statements and big price moves scroll across the bottom of the screen.

10 · TOOLS

# Flows, Curve and Scenario

## Flows



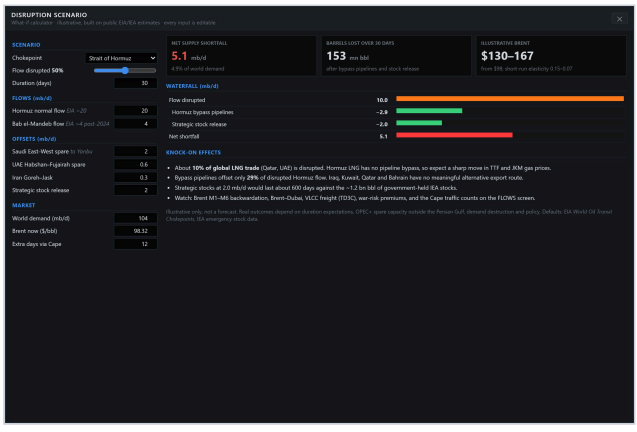

Ships per day through each chokepoint with a 7-day average (IMF PortWatch), and the terminal's own gate counts.

## Curve




Brent and WTI futures curves, now and a month ago, with each spread explained.

## Scenario

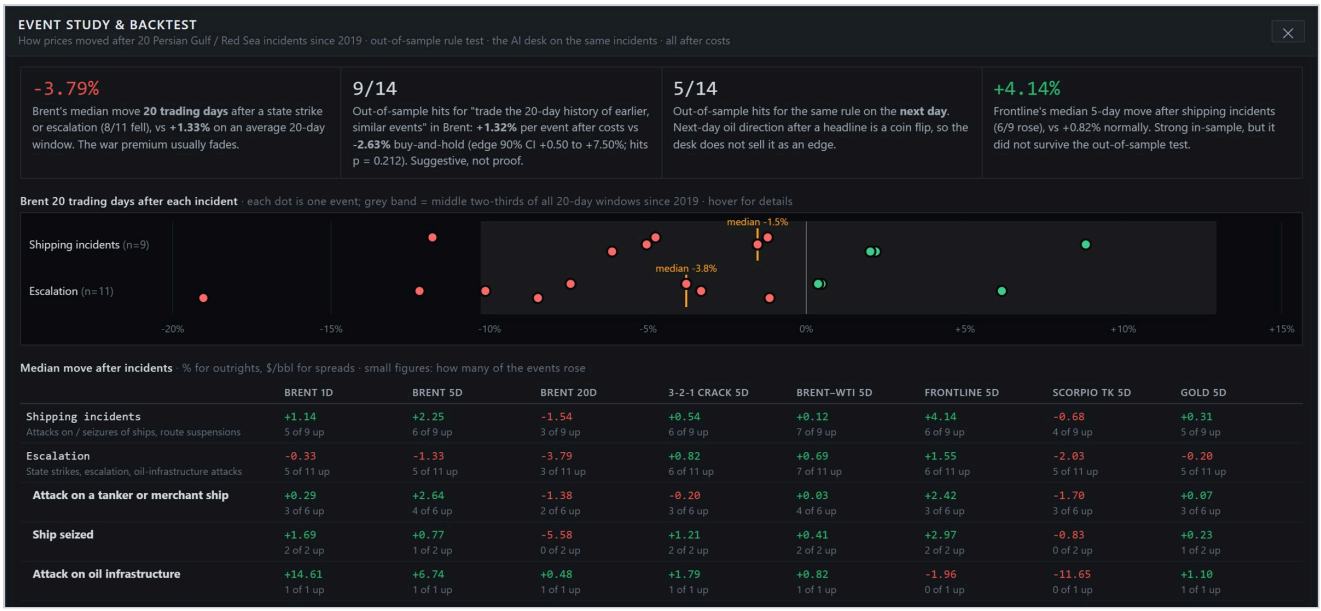



Choose a chokepoint, how much flow is lost and for how long; see the net supply shortfall and an illustrative price range.

> **Scenario figures are illustrative:** they use public estimates (US EIA, IEA) of flows, bypass pipelines and stocks, and are labelled as such in the tool.

# Research and Red Sea 2024

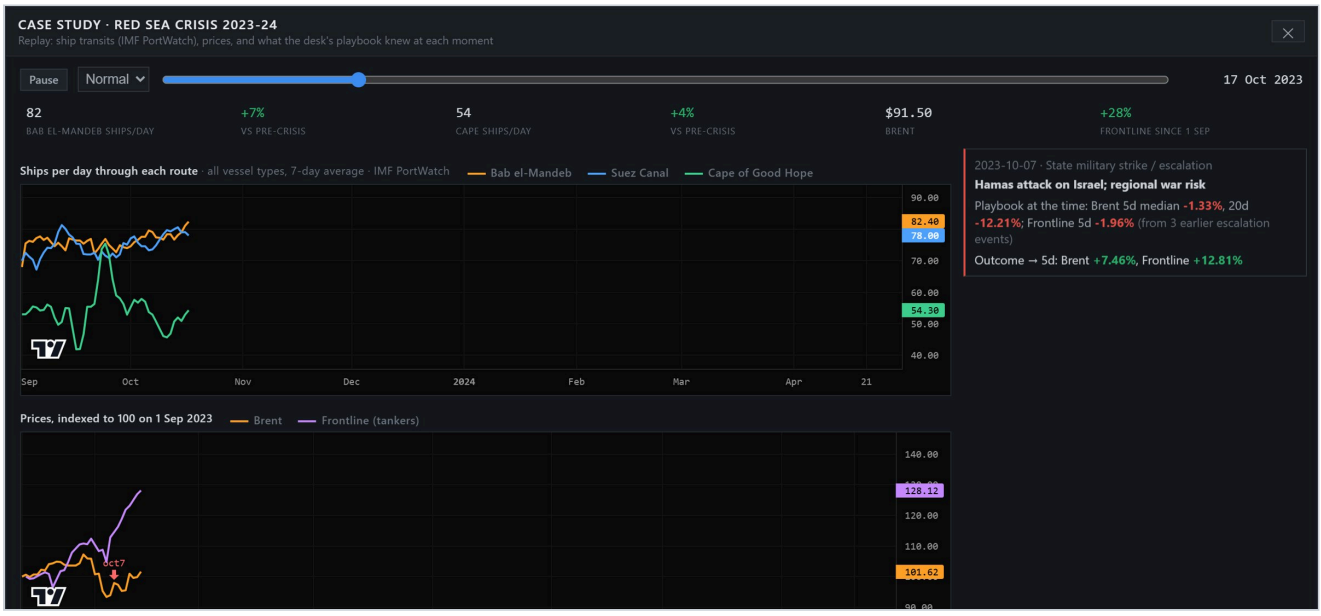
## Research




How oil and tanker shares moved after 20 Persian Gulf and Red Sea incidents since 2019, and how the rules and the AI desk did out of sample.

Every test is **walk-forward**: at each incident, a rule may only use incidents before it. Results are after trading costs, with 90% confidence intervals and p-values, and the page says plainly where an idea failed.

## Red Sea 2024




Press play to replay the 2023–24 crisis: ships per day through Bab el-Mandeb, Suez and the Cape, prices, and what the desk's playbook would have said.

12 · PHONE ALERTS, THE GUIDE AND QUICK REFERENCE

# Quick reference

## Phone alerts

Open **Alerts** in the tools menu. Install the free **ntfy** app and subscribe to the terminal's topic, or connect a **Telegram** bot; critical alerts and new trade proposals then arrive on your phone with a link back to the terminal. The same window shows the status of every data source.

## The guide

On your first visit a four-step tour highlights the risk banner, the maps, the trade desk and the evidence. Use the arrow keys or **Next**; **Esc** closes it. Reopen it at any time with **Guide** in the top bar.

## Everything at a glance

| To… | Do this |
|---|---|
| **See where risk is high** | Read the risk banner; hover a tile for the reasons. |
| **Jump to a waterway** | Click its risk tile, or its tab above the map. |
| **Find a ship from an alert** | Click LOCATE on the alert. |
| **Show only tankers** | Tick Tankers only in the top bar. |
| **Use VesselFinder's map** | Click Live map on the map panel. |
| **Replay the last 24 hours** | Click Replay on the map panel. |
| **Chart a price** | Click its tile in the price strip. |
| **Approve or reject a trade** | Trade desk tab, or click the desk tile in the banner. |
| **Stop all trading** | Kill switch (or Halt + flatten) on the trade desk. |
| **Ask a question** | Ask tab in the AI analyst panel. |
| **Filter official statements** | Official tab, then a side: Houthi, Iran, Israel, Persian Gulf, UK, UN or US. |
| **See the evidence** | Research, or Red Sea 2024, in the tools menu. |
| **Hear new alerts** | Tick Sound in the top bar. |
| **Close any window** | Press Esc or click ×. |

13 · EVIDENCE, LIMITATIONS AND DATA SOURCES

# Evidence and honest limits

## What the tests found

| Test | Result (out of sample, after costs) |
|---|---|
| **Next-day Brent direction** | 5 of 14 correct: no edge. |
| **Fading escalation, 20 days** | 9 of 14 correct; +1.3% per incident vs −2.6% buy-and-hold. Suggestive, not proof. |
| **AI desk (Gemini), 20 incidents** | 39 trades, 19 won, +0.14% of book: flat, no evidence of skill. |
| **Sealed hidden test (Agentverse)** | No better than buy-and-hold on 60 unseen real-price windows. |

These results are why the desk is built around human approval and hard limits rather than autonomy: the AI adds judgement, not a proven edge.

## Limitations

- Free AIS relies on shore receivers and covers little inside the straits; ships also go dark or spoof positions on purpose.
- Prices are delayed. Detections are leads to investigate, not confirmed events.
- The historical samples are small, and the AI may already know how past events ended; the live paper record is the only fully unseen test.

## Data sources

Yahoo Finance and CNBC (delayed prices) · IMF PortWatch (chokepoint transits) · aisstream.io (AIS) · US Treasury OFAC SDN list (sanctions) · named news outlets, Google News and Reddit (credited headlines) · official government and group channels · VesselFinder (embedded Live map) · Esri, GEBCO, NOAA and OpenSeaMap (map tiles) · US EIA and IEA (scenario defaults) · Google Gemini (analysis). Each is used under its provider's terms; see NOTICE.md in the repository.

GOWER STREET

## Take the next step.

| | |
|---|---|
| **Open the terminal** | gower-street-terminal-sg.onrender.com |
| **Source code** | github.com/itsaminzare/gower-street-terminal |
| **Built by** | Mohammadamin Zare, University College London |
| **Built for** | Agentverse 2026 · Finance Track |
| **AI** | Google Gemini (live analyst, trade desk and code contributions) |

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